There may, in specific instances, be additional first integrals present. For example, in a planar 2-body system in central forces, the angular momentum $xp_y - yp_x = l$ is a conserved quantity, thus $l \circ \mathscr{F}_t = l$.

Now if $I : \mathbb{R}^m \to \mathbb{R}$ has continuous partial derivatives, we may conclude directly that

$$I(\boldsymbol{a}) = I(\boldsymbol{b}) + \nabla I(z_*) \cdot (\boldsymbol{a} - \boldsymbol{b}),$$

where z_* is a point on the line in $\mathbb{R}^m$ connecting $\boldsymbol{a}$ to $\boldsymbol{b}$. Hence

$$|I(\boldsymbol{a}) - I(\boldsymbol{b})| \leq \|\nabla I(z_*)\| \|\boldsymbol{a} - \boldsymbol{b}\|.$$

This means if we know that $\|\nabla I(z)\|$ remains bounded, say less than B, in an open domain containing the solution, we could conclude that the error in I-values computed along a numerical trajectory is of order h^p,

$$|I(z(t_n)) - I(z_n)| \leq \frac{\bar{K}B}{2L} e^{nLh} h^p, \qquad h < \bar{h}, \tag{2.16}$$

with the same assumptions as are needed to characterize the convergence of the method. This also means that we can see a ready means of improving the error in a first integral: simply reduce the timestep. If we have identified a time interval of interest, $[0, \tau]$, then we will automatically have $nh < \tau$ and we need only to choose a stepsize h which brings the error within a target tolerance.

In many cases we will be satisfied with this level of control of the error in the first integrals, but there are a number of limitations of this approach. First, we may find that the constants involved, $\bar{K}$ and, more often, L, are simply so large that it is impossible to conclude anything useful from the bound given above. The stepsize would need to be so preposterously small to make the bound of practical interest that it is better to ignore the bound entirely.

Another limitation may arise when the time interval (the maximum of $\tau = vh$) is very long, as when we use trajectories to sample an energy landscape (the most common use of molecular dynamics). As this enters in the exponent in (2.16) it means that errors may continue to accumulate as we collect more samples and we would eventually leave the acceptable level of deviation in the first integral.

2.3 Geometric Integrators for Hamiltonian Systems

The discussion of the previous section suggests the need for methods with reliable conservation properties. We shall develop methods which, although their errors grow exponentially in time, nonetheless provide excellent energy conservation for very long times. These methods obtain their energy preservation properties indirectly: we design the methods to exactly conserve a certain geometric property